

Problem Set 6 Solutions

QTM 220

Due 1 December at 2:00pm ET on Canvas. Questions about submission requirements should be directed to your TA.

California CPS Revisited

- (a) Using the California sample of 25-35 year olds, regress income on education and age (in an additive model). Report your results in a table, including the standard errors that R automatically calculates.

```
cps<- read.csv("California_CPS_degrees.csv")

mod1<- lm(income~ education + age,data=cps)

summary(mod1)
```

Call:

```
lm(formula = income ~ education + age, data = cps)
```

Residuals:

Min	1Q	Median	3Q	Max
-226476	-62493	-20343	36193	752685

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-137806.3	24024.5	-5.736	1.1e-08 ***
education	16479.0	889.8	18.520	< 2e-16 ***
age	1051.6	694.4	1.514	0.13

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 101800 on 2243 degrees of freedom
Multiple R-squared: 0.1345, Adjusted R-squared: 0.1337
F-statistic: 174.2 on 2 and 2243 DF, p-value: < 2.2e-16

- (b) Create a matrix of your X information for your model (education and age) and include a column of 1s to estimate an intercept. Using the following matrix functions in base R, calculate $(X'X)^{-1}X'Y$ and confirm your resulting $\hat{\beta}$ s are the same as your lm results:

Let X be your matrix in R. Use `%*%` to multiply matrices, i.e. `X%*%X` will yield XX

Use `t(X)` to take the transpose of matrix X

Use `solve(X)` to take the inverse of matrix X

```
df.mat<- data.frame(constant=rep(1,length(cps$income)), edu=cps$education, age=cps$age)

X<- data.matrix(df.mat)

solve(t(X)%*%X)%*%t(X)%*%(cps$income)
```

```
      [,1]
constant -137806.340
edu       16478.997
age       1051.582
```

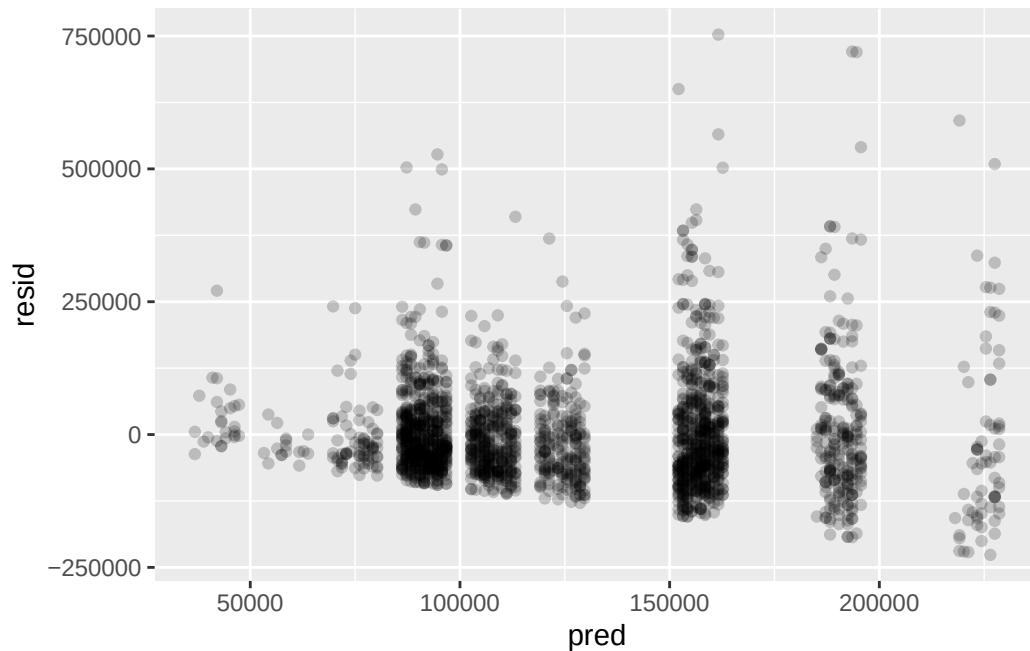
- (c) Extract the predicted values and residuals from your model using `predict(model name)` and `residuals(model name)`, then create a scatter plot of your predicted values and residuals. What do you see? Would you guess the errors are homoskedastic or heteroskedastic? Why?

```
pred<- predict(mod1)
resid<- residuals(mod1)

df<- data.frame(pred=pred,resid=resid)

rplot<- ggplot(df, aes(x=pred,y=resid)) + geom_point(alpha=.2)

rplot
```



The residuals are not equally dispersed across the predicted values of y . For high incomes, the residuals are much more spread out than the residuals for lower incomes. This suggests the errors are not constant (we use the residuals to make inferences about errors), suggesting heteroskedasticity. We shouldn't use the standard errors R gives us automatically because `lm` assumes homoskedastic errors.

- (d) Install the `sandwich` package and compute heteroskedasticity consistent standard errors for your model using `sandwich(model name)`.

```
sandwich(mod1)
```

	(Intercept)	education	age
(Intercept)	666941234	-15029237.613	-15481152.468
education	-15029238	1118548.385	2254.142
age	-15481152	2254.142	517429.329

```
sqrt(diag(sandwich(mod1)))
```

(Intercept)	education	age
25825.2054	1057.6145	719.3256

- (e) Compute the same HC standard errors for your model using your matrix X and our sandwich estimator,

$$\hat{V}[\hat{\beta}] = [X^T X]^{-1} X^T \text{diag}[(Y - X\hat{\beta})^2] X [X^T X]^{-1}.$$

```
D<- diag(resid^2)

dim(D)
```

```
[1] 2246 2246
```

```
betavar<- solve(t(X)%*%X)%*%t(X)%*%D%*%X%*%solve(t(X)%*%X)

sqrt(diag(betavar))
```

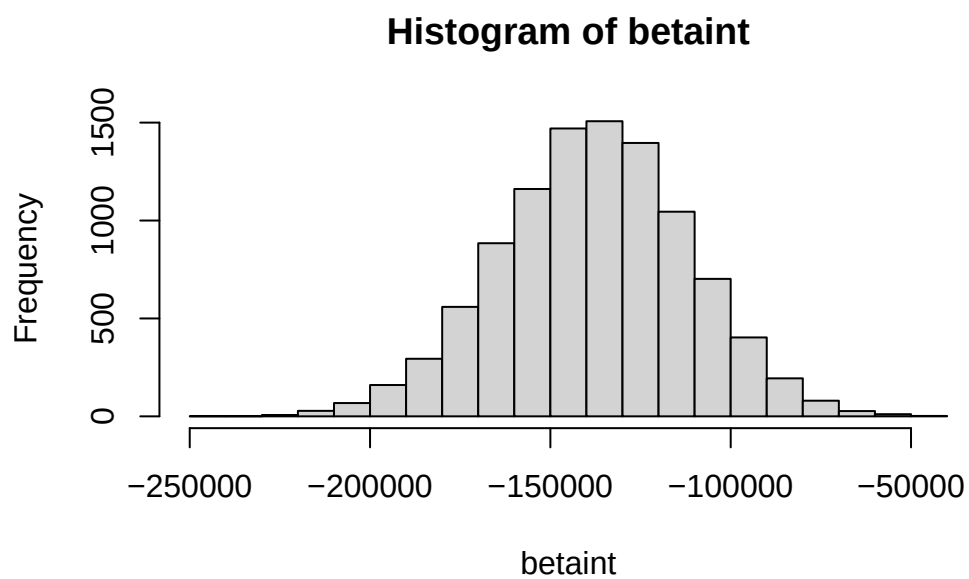
```
constant      edu      age
25825.2054  1057.6145   719.3256
```

- (f) Use a bootstrap to estimate the standard errors for each of the β s.

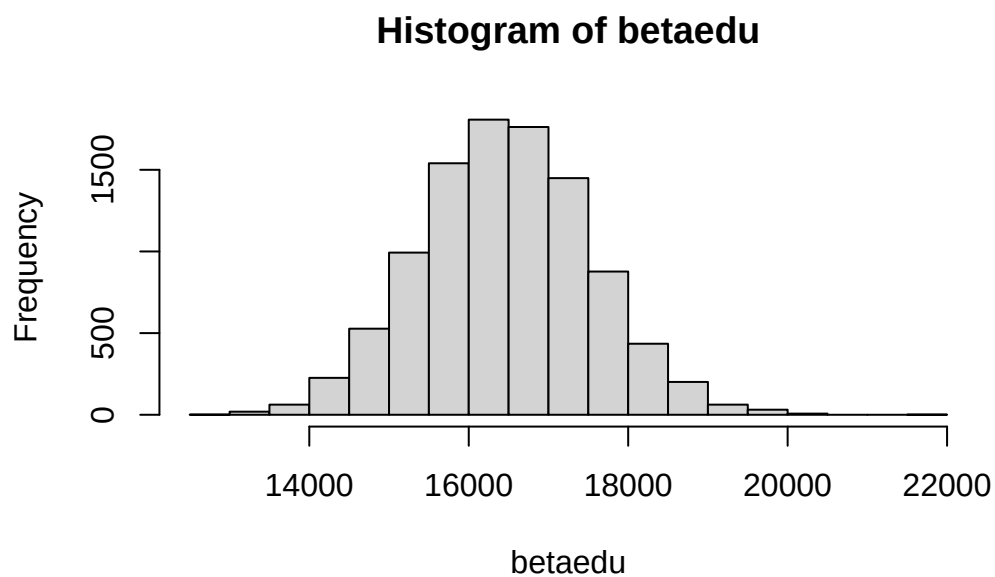
```
set.seed(220)
B=10000
betaint<- rep(NA,B)
betaedu<- rep(NA,B)
betaage<- rep(NA,B)
n=length(cps$income)

for( b in 1:B ){
  bootind = sample(1:n,size=n,replace=T)
  dfboot = cps[bootind,]
  mod<- lm(income ~ education+age, data=dfboot)
  betaint[b]<- mod$coef[1]
  betaedu[b]<- mod$coef[2]
  betaage[b]<- mod$coef[3]
}

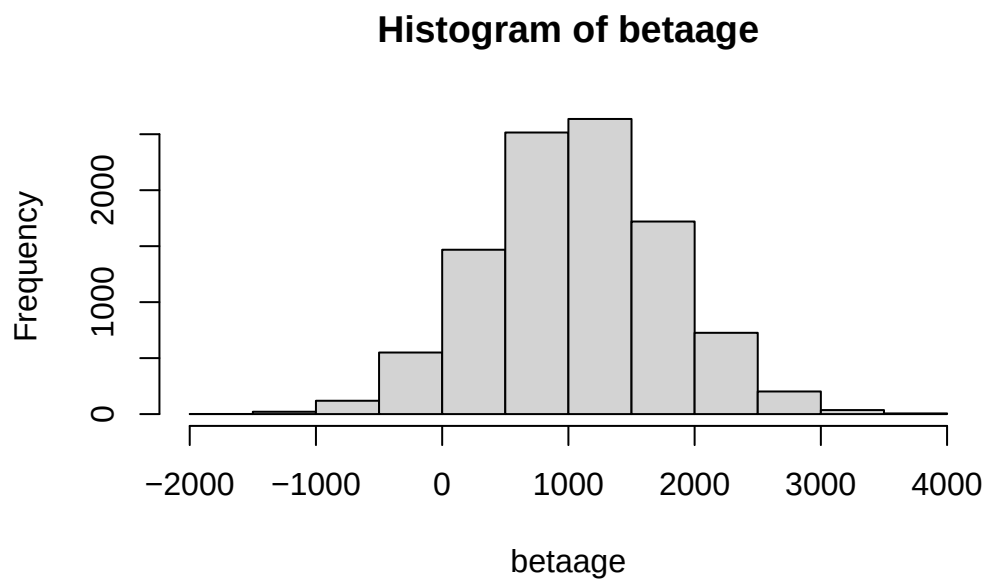
hist(betaint)
```



```
hist(betaedu)
```



```
hist(betaage)
```



```
sd(betaint)
```

```
[1] 26145.84
```

```
sd(betaedu)
```

```
[1] 1065.396
```

```
sd(betaage)
```

```
[1] 720.0392
```

```
quantile(betaedu,prob =c(.025,.975))
```

2.5% 97.5%
14412.29 18580.10

- (g) Compare the standard errors you got from R in part (a) to the robust (sandwich) you calculated in parts (e) and (f). Why are they different? How do these compare to what you estimated from the bootstrap?

The R standard errors that `lm` automatically gives us assume homoskedasticity, which we don't think is appropriate. Our bootstrapped errors are very similar to our heteroskedasticity-consistent sandwich standard errors. Note that the homoskedastic standard errors are lower, so we would be reporting greater precision/confidence than we ought to! Bad!

Causal Warm Up: Bone Density

Doctors (and researchers) hypothesize that higher calcium intake causes greater bone density (stronger bones). To test this, a clinic runs an experiment where patients are treated with a calcium supplement or a placebo for 1 year, after which their bone density is measured. Patients are first classified as old or young, then treatment assignment is randomized within the groups. Bone Density T-Scores indicate how “normal” someone's bone density is—0 is typical, positive numbers are better, negative numbers are worse (a threshold score of -2.5 is used to diagnose osteoporosis).

The resulting (fake) data is below.

Age Classification	Calcium Supplement	Bone Density T-Score
Old	1	-2
Old	0	-3
Old	1	-2
Old	0	0
Old	1	-1
Old	0	0
Old	0	-2
Old	0	0
Young	0	0
Young	1	1
Young	0	-1
Young	1	1.5

- (a) What is the probability of receiving treatment for the young? What is the probability of receiving treatment for the old?

Old: $\frac{3}{8}$; Young: $\frac{1}{2}$

- (b) Using simple difference in means, what is the treatment effect of the calcium supplement on bone density among the young? What is the treatment effect of the calcium supplement among the old?

Young: $\frac{1+1.5}{2} - \frac{0+-1}{2} = 1.75$

Old: $\frac{-2+-2+-1}{3} - \frac{-3+0+0+-2+0}{5} = \frac{-2}{3} = -0.667$

- (c) What is the average of these treatment effects, weighting by the number of participants in each category?

There are 4 young participants, so we weight the treatment effect of the calcium supplement on the young by 4/12; there are 8 old participants, so we weight the treatment effect of the calcium supplement on the old by 8/12

$$\frac{4}{12} * 1.75 + \frac{8}{12}(-0.667) = 0.13889$$

- (d) If you ignored the blocked randomization and calculated the ATE using the difference in means in treated and control, what would your estimate be?

$$\frac{-2+-2+-1+1+1.5}{5} - \frac{-3+0+0+-2+0+0+-1}{7} = -0.5 - -0.8571 = 0.3571$$

We would over estimate effect of a calcium supplement on bone density if we didn't take the blocking on age into account.

- (e) Input the data in R, creating a binary variable for age classification. Run two regressions: a simple additive of treatment on bone density score, and an interactive with age, treatment, and age*treatment. Write the equations for your two models using β s. Report your results in a table. Which model corresponds with which estimate of the treatment effect? Fill in the table below with the β s notation from your equations and the estimates from your model.

I am going to code old as 1, young as 0

```
age<- c(rep(1,8),rep(0,4))
treatment<- c(1,0,1,0,1,0,0,0,0,1,0,1)
bonedensity<- c(-2,-3,-2,0,-1,0,-2,0,0,1,-1,1.5)
df<- data.frame(age=age, treatment=treatment, bonedensity=bonedensity)

bonemod1<- lm(bonedensity~treatment,data=df)
summary(bonemod1)
```

Call:

```
lm(formula = bonedensity ~ treatment, data = df)
```


Residuals:

	Min	1Q	Median	3Q	Max
	-2.1429	-1.2321	0.3571	0.8571	2.0000

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.8571	0.5326	-1.609	0.139
treatment	0.3571	0.8251	0.433	0.674

Residual standard error: 1.409 on 10 degrees of freedom

Multiple R-squared: 0.01839, Adjusted R-squared: -0.07977

F-statistic: 0.1874 on 1 and 10 DF, p-value: 0.6743

```
bonemod2<- lm(bonedensity~treatment+ age + treatment:age,data=df)
summary(bonemod2)
```

Call:

```
lm(formula = bonedensity ~ treatment + age + treatment:age, data = df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2.000	-0.375	0.000	0.750	1.000

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.5000	0.7621	-0.656	0.530
treatment	1.7500	1.0777	1.624	0.143
age	-0.5000	0.9017	-0.555	0.594
treatment:age	-2.4167	1.3345	-1.811	0.108

Residual standard error: 1.078 on 8 degrees of freedom

Multiple R-squared: 0.5407, Adjusted R-squared: 0.3684

F-statistic: 3.139 on 3 and 8 DF, p-value: 0.08697

	β s
Avg Bone Density [Treated, Old]	
Avg Bone Density [Control, Old]	
Avg Bone Density [Treated, Young]	
Avg Bone Density [Control, Young]	
$\hat{ATE}_{Old}$	
$\hat{ATE}_{Young}$	

Neto and Cox

Use `netocox.csv` for Neto and Cox's 1997 data on the effect of runoff elections on the number of effective presidential candidates. The following variables are included:

- `RUNOFF` : whether the country has a runoff electoral system (treatment)
- `ENPRES` : the effective number of presidential candidates (outcome)
- `ENETH` : the effective number of ethnic groups

For the purposes of this exercise, let's assume that runoff systems are randomly assigned such that we can interpret estimates causally.

- Are the number of effective ethnic groups balanced across treatment and control countries? Make a (small) balance table reporting the means of the pre-treatment covariate.
- Fit an additive linear model of runoff and ethnic groups on number of presidential candidates. Calculate the Average Partial Derivative ($\hat{APD}$) of this model.

```
mod2<- lm(netocox$ENPRES~netocox$RUNOFF+netocox$ENETH)
summary(mod2)
```

Call:

```
lm(formula = netocox$ENPRES ~ netocox$RUNOFF + netocox$ENETH)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.3094	-1.0066	-0.2142	0.9920	1.9826

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.2629	0.8691	2.604	0.0219 *
netocox\$RUNOFF	0.6311	0.6087	1.037	0.3187

```
netocox$ENETH    0.3661    0.4986    0.734    0.4759
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1.211 on 13 degrees of freedom
```

```
Multiple R-squared:  0.1206,    Adjusted R-squared:  -0.01464
```

```
F-statistic: 0.8918 on 2 and 13 DF,  p-value: 0.4336
```

- (c) Fit an interactive linear model of runoff and ethnic groups on number of presidential candidates. Calculate the Average Partial Derivative ($\hat{APD}$) of this model.
- (d) Using the interactive model, impute $Y(1)$ s for the no runoff countries and $Y(0)$ s for the runoff countries. Fill in your estimates in the table below, then calculate the ATE of runoff on presidential candidates.

Country	Runoff?	$Y(0)$	$Y(1)$
Argentina	0		
Austria	1		
Brazil	1		
Colombia	0		
Costa Rica	1		
Dominican Republic	0		
El Salvador	1		
Ecuador	1		
Finland	0		
France	1		
Iceland	0		
Korea	0		
Peru	1		
Portugal	1		
USA	0		
Venezuela	0		

- (e) Fit an interactive polynomial model of runoff and ethnic groups (and ethnic groups squared) on number of presidential candidates. Calculate the Average Partial Derivative ($\hat{APD}$) of this model.
- (f) Using the interactive polynomial model, impute $Y(1)$ s for the no runoff countries and $Y(0)$ s for the runoff countries. Fill in your estimates in the table below, then calculate the ATE of runoff on presidential candidates.

Country	Runoff?	Y(0)	Y(1)
Argentina	0		
Austria	1		
Brazil	1		
Colombia	0		
Costa Rica	1		
Dominican Republic	0		
El Salvador	1		
Ecuador	1		
Finland	0		
France	1		
Iceland	0		
Korea	0		
Peru	1		
Portugal	1		
USA	0		
Venezuela	0		

Menopausal Hormone Therapy

Review the WHI/WHI(MS) studies as well as the Kim et al. 2021 study on neurodegenerative diseases.

- (a) Fill in the following table comparing WHI and Kim et al.

	WHI Study	Kim et al.
Method/approach used		
Population/participants		
Are there relevant populations not included? Who?		
Treatment(s)		
Outcome(s)		

- (b) What is a potential omitted variable from the Kim et al. study? Does omitting that variable bias their estimate of the treatment effect of hormone therapy? Use the formulae from class to discuss whether you think their estimate might be too high, too low, or accurate.
- (c) Is there a concern about omitted variable bias in the WHI? Why or why not? Use the formulae from class to discuss whether you think their estimate might be too high, too low, or accurate.

Review the Estrogens General Statement clinical reference and read the recent NYT magazine piece about menopausal hormone therapy, the WHI's response to the NYT piece, and JoAnn Manson (WHI Principal Investigator) and Andrew M. Kaunitz's perspective on menopause management (posted on Canvas).

- (a) How do you assess the medical guidelines? Are the certainties and uncertainties of the body of research communicated effectively? Based on what you have read and our class discussions, who should be prescribed hormone therapy? Who should not be prescribed hormone therapy? Discuss patient characteristics including age, pre-existing conditions, vasomotor symptoms, and whatever else you deem relevant, citing the studies we have worked with this semester as needed.

- (b) How did making patient recommendations in the previous part feel? What aspects of our readings/case study did you find most convincing and why? What incentives do researchers and pharmaceutical companies have in research on hormone therapy medications? What incentives do doctors/providers have when making recommendations to their patients? How do you think these incentives affect the accumulation of knowledge and patient outcomes?